SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

C01_AT2 — DATA INTERPRETATION EXERCISE

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO1 –Apply (BL3)	**Assessment:**	Assessment Tool 2 – Data Interpretation
Weightage:	20% of CIA	**Total Marks:**	20(2 Questions × 20 Marks)
CO1 Statement:	*Understand the fundamental concepts, principles, and techniques of data visualization. (BL2)*		
Student Name:	PRIYANKA.G	**Reg. No.:**	192524124
Section / Batch:		**Date:**	28.07.2026

Code of Conduct

I Priyanka. G /192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.

Annexure A – DATA INTERPRETATION

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Problem 10 (10 Marks)

Scenario: A market share pie chart shows four companies with shares of 45%, 30%, 15%, and 10%.

a) What coordinate system underlies a pie chart, and what two variables (r, θ) define each slice's boundary?

Answer:

A pie chart is built on the polar coordinate system, in which every point is located by a radius (r) and an angle (θ) measured from a fixed centre, rather than by horizontal and vertical (x, y) positions as in Cartesian coordinates. In this chart, the radius (r) is held constant for every slice, since it simply defines the fixed outer boundary of the circle and does not encode any data value. The angle (θ), on the other hand, is the variable that actually carries information: each category's proportion of the whole is mapped to a proportional angular sweep around the centre, so the boundary of every slice is defined by the two radii drawn at its starting and ending angles, together with the arc that connects them. In this way, the numerical (proportion) data is encoded through the aesthetic mapping of value → angle, while the nominal data (company names) is encoded through the aesthetic mapping of category → colour hue, as seen in the legend (Company A, B, C, D).

b) Calculate the angle (in degrees) of the slice representing the 30% market share.

Answer:

A complete pie chart corresponds to a full circle of 360°, which represents 100% of the market. Since the angle of a slice is directly proportional to its share of the whole, the angle for any category can be found using the formula: Angle = (Percentage share ÷ 100) × 360°. For Company B, which holds a 30% market share, the calculation is: Angle = (30 ÷ 100) × 360° = 0.30 × 360° = 108°.

Therefore, the slice representing Company B's 30% market share occupies an angular sweep of 108° in the pie chart, which is visibly consistent with the chart shown in Figure 1, where Company B's slice is clearly narrower than Company A's (45%, i.e., 162°) but wider than Company C's (15%, i.e., 54°).

c) What is the key limitation of polar coordinates (pie charts) when more than 5–6 categories are shown?

Answer:

The key limitation is that human perception is far less accurate at comparing angles and areas than it is at comparing lengths or positions along a common scale, as established in graphical-perception research (e.g., Cleveland and McGill's hierarchy of elementary perceptual tasks). When a pie chart contains more than 5–6 categories, several slices inevitably become small and similar in angular size, making it very difficult for the viewer to judge, at a glance, which slice is genuinely larger or by how much. This is compounded by practical issues such as an overcrowded colour legend, labels that overlap or must be pulled outside the chart, and the use of near-identical hues to distinguish many categories, all of which increase visual clutter. As a result, for datasets with many categories, alternative representations such as a bar chart — which relies on the more accurately perceived cue of length along a common baseline — are generally recommended over a pie chart.

Data Visualization

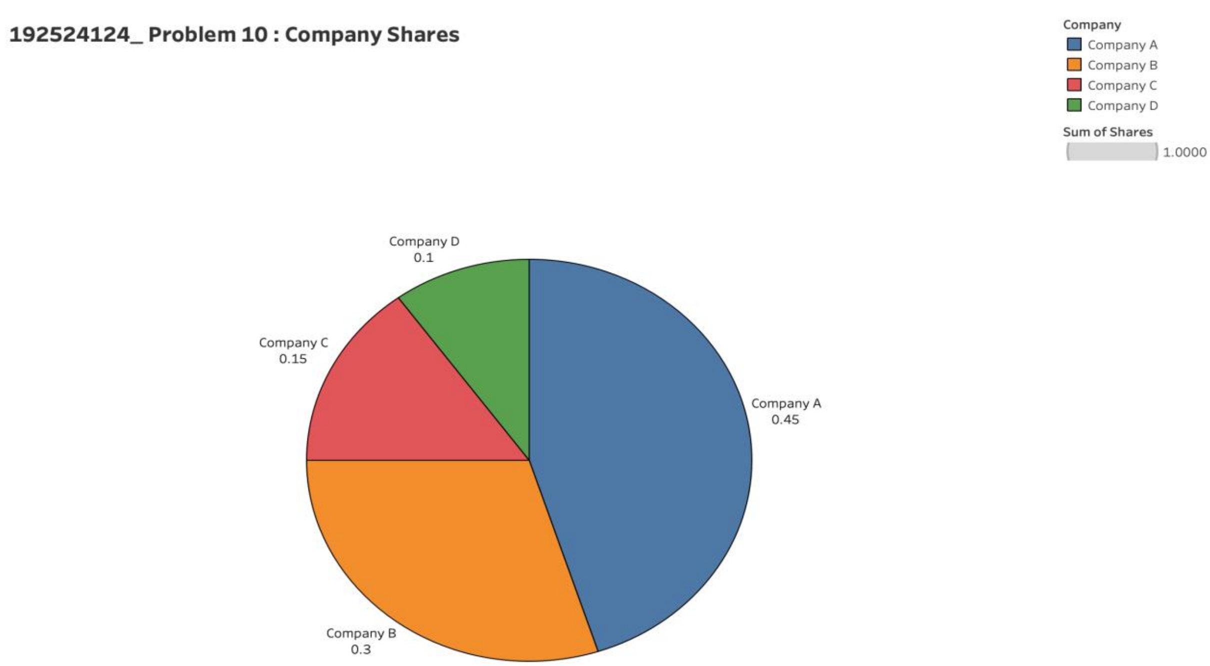

Figure 1: Market Share Distribution of Companies A, B, C, and D Pie Chart

Problem 20 (10 Marks)

Scenario: A temperature anomaly map (showing how much warmer or cooler each region is compared to a historical average) uses a diverging red–blue colour scale.

a) What does the color white (or the midpoint color) represent on this map?

Answer:

In a diverging colour scale, the midpoint colour — white, in this case — represents the neutral reference value of zero anomaly, meaning no deviation from the historical average temperature. It marks the boundary between the two halves of the scale: values above this midpoint are mapped through increasingly saturated red, and values below it are mapped through increasingly saturated blue. In the map shown in Figure 2, the Central region (whose bar is not shown, i.e., at or near 0°C) would sit at or close to this white midpoint, indicating that its temperature is essentially the same as its historical average, with neither noticeable warming nor cooling.

b) If a region is shown in deep red, what can you conclude about that region's temperature relative to the historical average?

Answer:

A deep red colour indicates a large positive temperature anomaly, meaning that the region's current temperature is substantially higher than its historical average, i.e., it has experienced significant warming. The intensity (saturation) of the red is itself meaningful: the deeper or more saturated the red, the greater the magnitude of the positive deviation. In Figure 2, the South region is shown in deep red at an anomaly of +5.0°C, indicating that it is markedly warmer than its historical baseline, in clear contrast to the North region, which is shown in deep blue at approximately −4.0°C, indicating a region that is markedly cooler than its historical average.

c) Why would a sequential (single-hue) color scale be a poor choice for this particular dataset?

Answer:

A sequential (single-hue) colour scale encodes only the magnitude of a value through variation in lightness or saturation of one colour, running from a light shade (low value) to a dark shade (high value) of the same hue. It works well for data that increases steadily from a natural minimum, such as population count or rainfall. Temperature anomaly data, however, is inherently diverging: it has a meaningful, natural midpoint at zero (no change from the historical average), and values can be either positive (warmer) or negative (cooler) relative to that midpoint. A single-hue sequential scale has no way to distinguish direction — it can only show 'more' or 'less' of one hue, so a strongly negative anomaly and a strongly positive anomaly could easily be confused, or the critical zero point would be lost entirely. A diverging red–blue scale, in contrast, explicitly encodes both the direction (red for warming, blue for cooling) and the magnitude (colour intensity) of the anomaly, which is essential for correctly interpreting this dataset.

Data Visualization

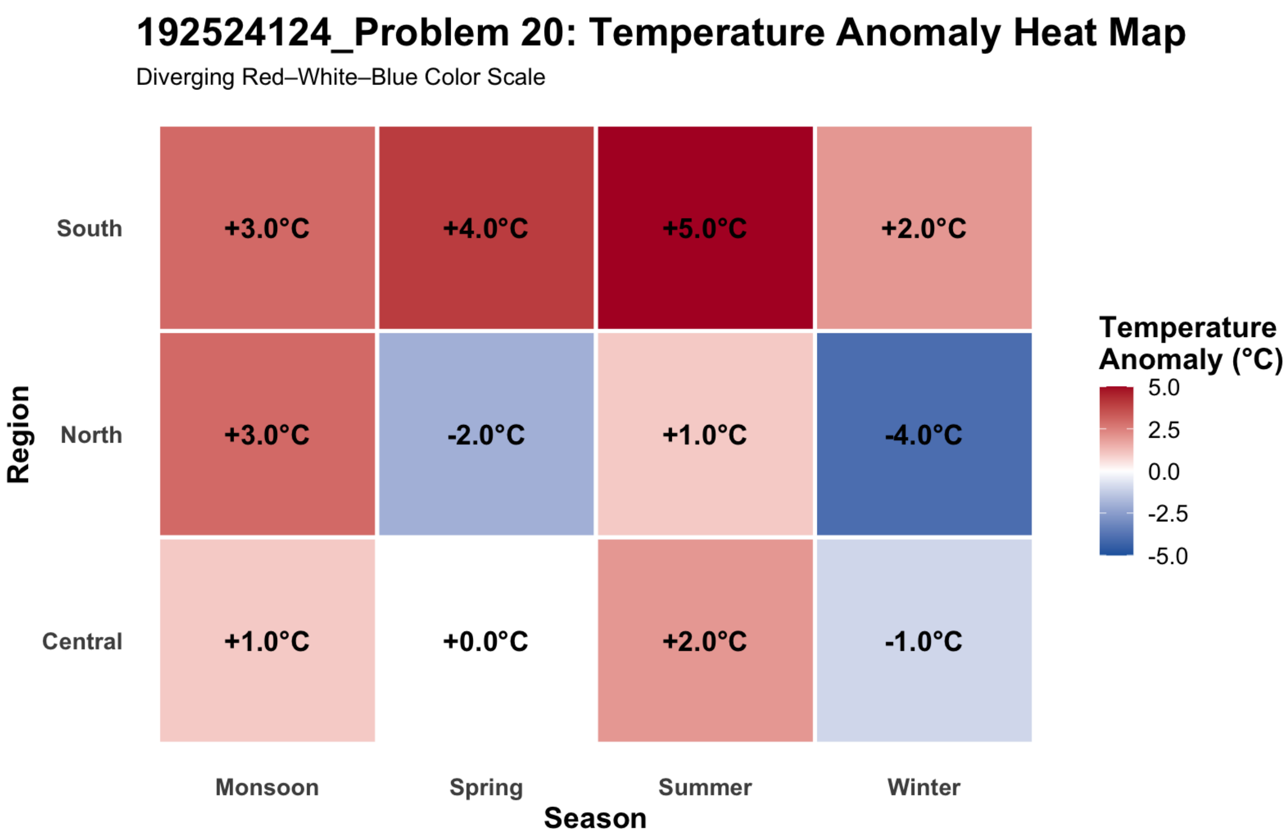

Figure 2: Temperature Anomaly Haet Map (Central, North, South)